



Shahid Beheshti University

Machine Learning

Assignment 5

Theoretical Section

- Question 1:** What are some limitations of K-Means Clustering? Describe common methods like random initialization and K-Means++ for initializing centroids. Can K-Means be used with categorical data? Justify your answer.
- Question 2:** You applied the DBSCAN algorithm on a dataset and noticed that it produced too many small clusters, and a large number of points were labeled as noise. What might be the cause of this behavior in terms of DBSCAN's parameters?
- Question 3:** a) Why can GMM be considered a soft clustering method? How does this differ from hard clustering methods?
b) Can clustering be used for anomaly detection? Explain how DBSCAN or GMM can detect outliers.
- Question 4:** a) Explain the difference between the silhouette score and the Davies-Bouldin index. Can we use the silhouette score for a dataset with one cluster? Explain why.
b) Specify how the silhouette score changes for well-separated clusters versus clusters that overlap.
- Question 5:** a) Compare and contrast the usage of PCA and T-SNE. In what scenarios is using T-SNE more advantageous?
b) Does it make sense to chain two different dimensionality reduction algorithms?
c) If the data are time series, which distance metric is more appropriate for measuring the distance between samples in clustering, and why?

Question 6: The table below records the scores of five students in three subjects. Ignore the **Student ID** column and perform Principal Component Analysis (PCA), retaining two principal components. This should include centering the data, calculating the covariance matrix, finding the eigenvectors, and finding the two principal components.

Student	Math	English	Art
1	90	60	90
2	90	90	30
3	60	60	60

Question 7: How can clustering be used to improve the accuracy of the linear regression model?

Question 8: In hierarchical clustering, if a dendrogram shows that almost all merges occur at approximately the same height, what does this indicate about the underlying data? Additionally, explain how using a centroid-based linkage can lead to distance inversion in the dendrogram.

Practical Assignment: Personality Profiling with Clustering

Objective

The objective of this assignment is to apply various unsupervised clustering algorithms to the "Big Five Personality Test" dataset to identify distinct personality profiles among individuals. You will evaluate the performance of these clustering models using appropriate metrics and visualize the resulting clusters to gain insights into human personality traits.

Dataset Overview

The ["Big Five Personality Test"] dataset, available on Kaggle, comprises responses from individuals on a personality assessment. The features are questions related to each personality type and the values are scores given to them by participants from 1 to 5. Personality types are as below:

- Openness to Experience (OPN): Measures inventiveness and curiosity vs. consistency and caution.
- Conscientiousness (CSN): Measures efficiency and organization vs. extravagance and carelessness.

- Extraversion (EXT): Measures outgoingness and energetic behavior vs. solitude and reservedness.
- Agreeableness (AGR): Measures friendliness and compassion vs. challenging and detached behavior.
- Neuroticism (EST): Measures emotional stability and calm vs. nervousness and moodiness.

Each feature's name indicates which personality the question is related to, by a key (OPN, CSN, EXT, AGR, EST). The dataset may also contain demographic information or other related features that could be relevant for clustering. Your task is to use the personality trait scores to group individuals into meaningful clusters.

Assignment Tasks

1. Data Preprocessing and Exploratory Data Analysis (EDA)

Load and Clean: Load the dataset into your environment. Address any missing values (e.g., imputation, removal) and identify/handle potential outliers.

Feature Selection: Identify the columns corresponding to the Big Five personality traits (and potentially other relevant numeric features if applicable) that will be used for your clustering.

Descriptive Statistics & Visualization:

- You can compute descriptive statistics (mean, median, standard deviation, etc.) for each personality trait.
- Visualize the distributions of individual personality traits (e.g., histograms, box plots).
- Analyze the correlations between the different personality traits.

Dimensionality Reduction: Consider applying dimensionality reduction techniques like Principal Component Analysis (PCA) or t-Distributed Stochastic Neighbor Embedding (t-SNE) to visualize the data in lower dimensions before clustering, to get a sense of inherent groupings.

Scaling: Normalize or standardize the numerical features. Explain why scaling is important for distance-based clustering algorithms.

2. Clustering Model Development and Evaluation

Apply various unsupervised clustering algorithms to your preprocessed data.

Clustering Models to Implement:

- **K-Means Clustering:**
 - Experiment with different values for 'k' (the number of clusters).

- Use techniques like the Elbow Method (using inertia/distortion) and the Silhouette Score to help determine the optimal number of clusters.
- **Hierarchical Clustering:**
 - Experiment with different linkage methods (e.g., Ward, Complete, Average).
 - Visualize the dendrogram to aid in determining the optimal number of clusters.
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):**
 - Experiment with different eps and min_samples parameters to get a better result.
 - Discuss the implications of noise points identified by DBSCAN.

Evaluation Metrics:

For each clustering model and chosen number of clusters, report the following internal validation metrics:

- **Silhouette Score:** Measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation). A higher score indicates better-defined clusters.
- **Davies-Bouldin Index:** Measures the average similarity ratio of each cluster with its most similar cluster. A lower score indicates better partitioning.

Report each metric for all models and their chosen parameters/number of clusters.

3. Interpretation & Visualization

Cluster Visualization:

- Visualize the clusters in 2D or 3D plots, coloring data points by their assigned cluster.

Insights and Interpretation:

- Based on your visualizations and the average trait scores, interpret what defines each cluster.
- Compare the results from different clustering algorithms and discuss which algorithm performed best.

Submission Guidelines

Your final submission must include:

1. A Jupyter Notebook with all code, outputs, visualizations, and markdown explanations clearly presented.
2. A well-documented PDF report summarizing your methodology, findings, interpretations, and conclusions for each task.

If you would like to take the test yourself, here is the [\[link\]](#) :)